

SARDAR PATEL UNIVERSITY
BCA (Semester-VI) Examination April - 2022
US06EBCA01 (Information Security)
Date : 07 - 04 - 2022, Time : 12:00 p.m to 02:00 p.m



Total Marks : 70

Q.1 Multiple Choice Questions.**[10]**

- 1 Which action is that compromises the security of information owned by an organization?
a) Threat b) Security Attack c) Service attack d) None of these
- 2 _____ the process that is designed to detect, prevent or recover from a security.
a) Security mechanism b) Security Attack c) Service attack d) Threat
- 3 A threat which intercept or modify data on behalf of users who should not have access to that data is _____.
a) Information access threats b) Service threats c) Access Control d) NonRepudiation.
- 4 Cryptography is also known as _____.
a) plain text writing b) stream writing c) Secret writing d) None of above
- 5 _____ is the original message or data that is fed into the algorithm as input.
a) plaintext b) Encryption algorithm c) secret key d) ciphertext
- 6 _____ performs various substitutions and transformations on the plaintext.
a) plaintext b) Encryption algorithm c) secret key d) ciphertext
- 7 A _____ is a program or a part of program that are activated by a trigger.
a) Virus b) Logic bomb c) Worms d) Trojan horses
- 8 What the virus does besides spreading is known as _____.
a) Triggering b) Dormant c) Infection vector d) Payload
- 9 The _____ server in the DNS needs only list a limited number of public IP addresses.
a) DNS b) NAT c) Proxy d) Firewall
- 10 _____ Server is a server that creates a secure tunnel connection.
a) DNS b) VPN c) Tunnel d) authentication

Q.2 Fill in the Blanks and True or False.**[08]**

1. _____ Attacks are in nature of eavesdropping on, or monitoring of transmission.
2. The process of attempting to discover the plaintext or key is known as _____.
3. IDS stand for _____.
4. _____ Controls access to service according to which user is attempt.
5. Authentication is one type of Security services.(True/False)
6. Block cipher process the input elements continuously, producing output one element at a time.(True/False)
7. A Virus is a program or a part of program that are activated by a trigger. (True/False)
8. Behavior controls how particular services are used.(True/False)

Q.3 Answer the Following.(Any 10)**[20]**

1. List the different types of security attacks.
2. List the different types of security services.
3. What do you mean by security mechanism and security services?
4. Define plain text and cipher text.
5. Differentiate public key and private key.
6. Define cryptography and crypto analysis.
7. Define: Virus and Worm
8. Write the three parts of computer virus.
9. Write the approach used by the antivirus.
10. Define Digital signature.
11. What is circuit level gateway?

12. List four techniques used by firewall.

Q.4 Answer the Following. (Any 4)

1. What is security attack? Explain in detail with its types?
2. Explain the network security model in detail?
3. What is cryptography? Explain symmetric key and asymmetric key cryptography.
4. Explain the symmetric encryption principles and model of symmetric encryption.
5. Explain different nature of viruses in detail.
6. Write a note on Intruders and Backdoor.
7. What is firewall? Explain characteristics of firewall in detail with Goal, Scope, technique and limitation.
8. Write comparison between conventional and digital signature.

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